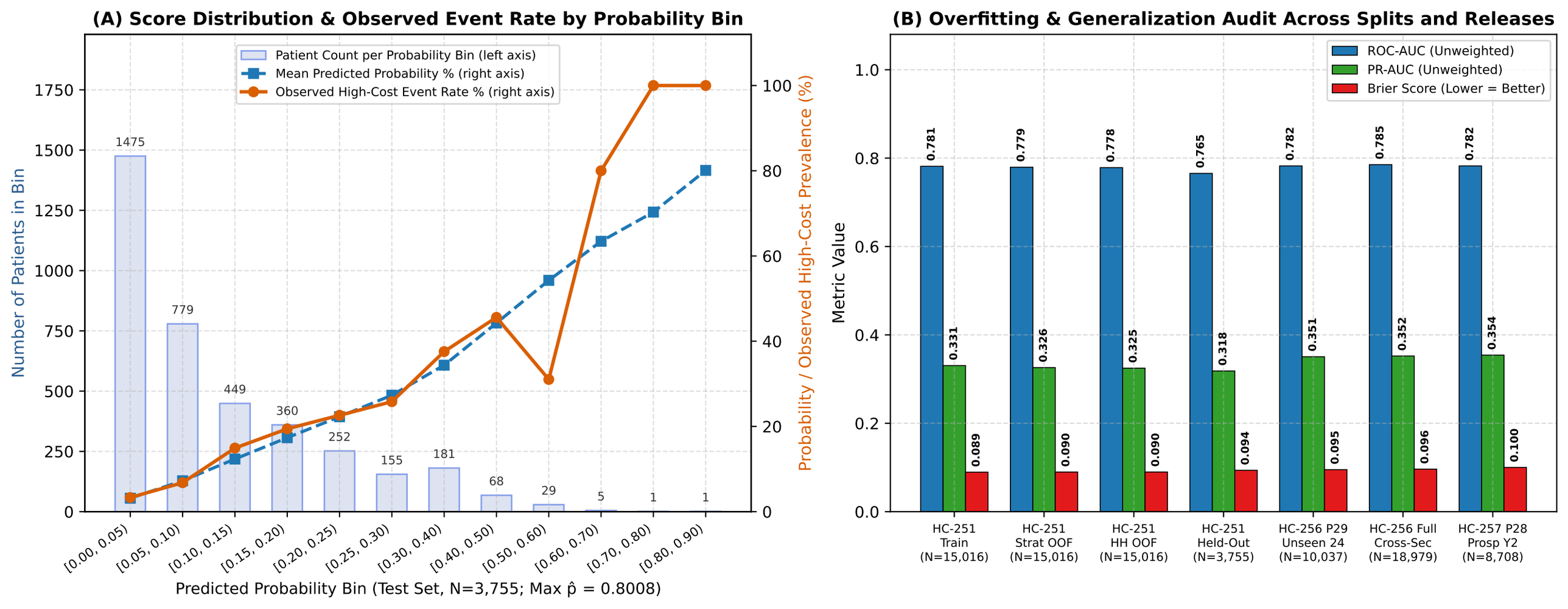
(A) Score Distribution & Observed Event Rate by Probability Bin
Patient Count per Probability Bin (left axis)
Mean Predicted Probability % (right axis)
Observed High-Cost Event Rate % (right axis)
Number of Patients in Bin
Probability / Observed High-Cost Prevalence (%)
1475
779
449
360
252
155
181
68
29
5
1
1
[0.00, 0.05)
[0.05, 0.10)
[0.10, 0.15)
[0.15, 0.20)
[0.20, 0.25)
[0.25, 0.30)
[0.30, 0.40)
[0.40, 0.50)
[0.50, 0.60)
[0.60, 0.70)
[0.70, 0.80)
[0.80, 0.90)
Predicted Probability Bin (Test Set, N=3,755; Max p̂ = 0.8008)
(B) Overfitting & Generalization Audit Across Splits and Releases
ROC-AUC (Unweighted)
PR-AUC (Unweighted)
Brier Score (Lower = Better)
Metric Value
0.781
0.331
0.089
0.779
0.326
0.090
0.778
0.325
0.090
0.765
0.318
0.094
0.782
0.351
0.095
0.785
0.352
0.096
0.782
0.354
0.100
HC-251 Train (N=15,016)
HC-251 Strat OOF (N=15,016)
HC-251 HH OOF (N=15,016)
HC-251 Held-Out (N=3,755)
HC-256 P29 Unseen 24 (N=10,037)
HC-256 Full Cross-Sec (N=18,979)
HC-257 P28 Prosp Y2 (N=8,708)